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GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation

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***Abstract*—In-hand manipulation is an essential capability for robotic hands, enabling complex object interactions such as grasping, repositioning, and lifting. This paper presents GLINT (Grasp-Lift Idle-stroke Nonlinear Transmission) hand, a novel grasp-and-lift hand realized through a lost-motion transmission mechanism. The design introduces a dual-phase operation: in the idle-stroke phase, the hand achieves grasping by allowing free alignment without force output; in the engaged phase, it performs lifting by transmitting force effectively. This mechanical switching enables an inherent form of in-hand manipulation, where the grasp-to-lift transition is achieved without relying heavily on upper-level controllers. A kinematic and dynamic model is developed to describe the switching conditions and the force–displacement relationship. The design is implemented and experimentally validated on a robotic platform, demonstrating accurate and reliable grasping and lifting with reduced control requirements and lower energy consumption. GLINT hand highlights a new paradigm for in-hand manipulation and offers potential applications in aerial manipulation, warehouse automation, and lightweight robotic grippers.**

Keywords—Idle-stroke, Grasp-and-lift hand, In-hand manipulation, Robotic lifting mechanism, Warehouse automation

I. Introduction

Object grasping and lifting remain central challenges in robotics, automation, and artificial intelligence, especially in unstructured and dynamic environments that demand robustness, efficiency, and simplicity. Conventional robotic systems typically rely on tightly coupled perception–planning–control stacks to achieve reliable manipulation. While effective, these pipelines incur substantial computational, communication, and energy costs. To mitigate such burdens, a growing body of work embeds manipulation logic within the end-effector itself, shifting capability from software to hardware.

In-hand manipulation has emerged as a promising paradigm. Weinberg *et al.* [1] surveyed reinforcement learning and data-driven approaches that advanced dexterous in-hand control. Complementing learning-based methods, multimodal sensing integration has become a key driver of progress. For instance, Yuan *et al.* [2] developed a point-cloud-level visuo-tactile fusion method enabling dexterous object rotation, and Huang *et al.* [3] combined dense tactile sensing with vision to refine fine-grained manipulation. Li *et al.* [4] introduced predictive force attention for adaptive visuo-tactile fusion, dynamically rebalancing sensory contributions during contact.

Other frameworks continue to demonstrate the promise of visuo-tactile pipelines. Tian *et al.* [5] applied soft-fusion contrastive learning to improve multimodal representations. Tu *et al.* [6] combined tactile-visual fusion with SelectLSTM for robust in-hand pose estimation. Chen *et al.* [7] proposed visuo-tactile transformers to unify cross-modal features, while Bauza *et al.* [8] achieved precise pick-and-place through simulation-trained visuotactile learning. Expanding this field, Suresh *et al.* [9] introduced NeuralFeels with neural fields for online shape and pose estimation in multi-fingered manipulation, and Mao *et al.* [10] showed how tactile-vision fusion improves recognition, grasping, and control robustness.

On the hardware side, tactile sensing plays an equally critical role. James and Lepora [11] demonstrated how slip detection stabilizes multi-fingered grasping. Deng *et al.* [12] applied tactile force feedback for adaptive pressure control. Li *et al.* [13] reviewed tactile perception as a foundation for manipulation, and Li *et al.* [14] provided a comprehensive overview of robotic intelligent grasping, emphasizing tactile integration in grasp planning. Further, Fan *et al.* [15] introduced the CrystalTac family, a set of vision-based tactile sensors manufactured via rapid monolithic processes, underscoring the importance of hardware-level sensing innovations. Collectively, these works converge on the insight that perception fusion and contact feedback are indispensable for advancing manipulation capability.

In parallel with sensing-driven advances, researchers have pursued mechanism-level innovations to offload part of the control burden into hardware. Ding *et al.* [16] introduced the Semi-Peaucellier linkage integrated with a differential mechanism, reported in their SP-Diff gripper, which achieved hybrid switching between linear parallel pinching and adaptive enveloping grasping through a Peaucellier-derived structure. This work illustrates how classical kinematic topologies can be reconfigured to improve adaptability and reduce system-level dependence on perception and planning.

Building on such insights, the present study proposes a novel idle-stroke-based grasp-and-lift hand as shown in Fig. 1. Unlike adaptive pinching structures, the proposed design embeds grasp-to-lift phase switching directly into the

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transmission system. During the idle-stroke phase, the mechanism allows free alignment without transmitting force, realizing grasping at the structural level. Once engaged, the system transitions into a force-transmitting phase for lifting. By encoding the phase-switching logic within the mechanical architecture, this approach reduces reliance on high-level control, offering robustness under perceptual delays, limited communication, or resource-constrained conditions.

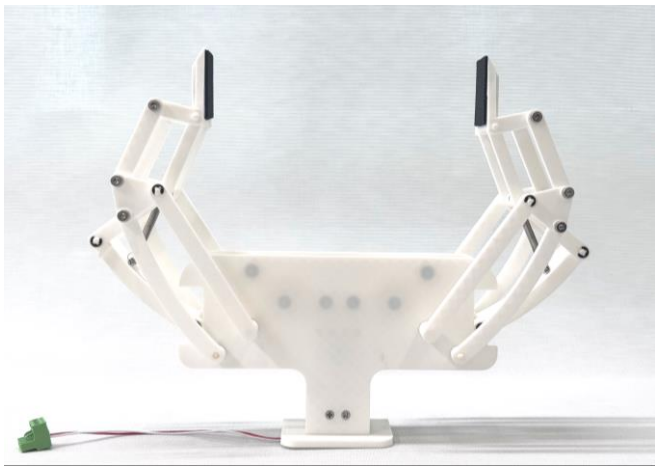

Fig. 1. Prototype of the proposed GLINT hand.

We fabricate a physical prototype of the idle-stroke hand, derive its kinematic and dynamic models to characterize grasp-to-lift transitions and force–displacement behavior, and validate performance through real-world experiments. Results confirm smooth phase switching, accurate force transmission, and efficient energy use. The remainder of this paper is organized as follows: Section II presents the design principles; Section III develops the kinematic and dynamic analyses of the proposed mechanism; Section IV reports the hardware experiments; and Section V concludes with discussions and future directions.

II. Design Concept

A. Fingertip Translation with Dual–Parallelogram

To maintain fingertip orientation while translating toward the object, the hand uses a cascaded dual-parallelogram linkage (Fig. 2a). The cascaded layout conveys input rotation to an approximately pure translational motion of the distal pad, preserving the local contact normal and reducing tangential disturbances during approach. In practice, the near-prismatic behavior holds over the usable stroke with sub-degree tip rotation under standard machining tolerances, which mitigates lateral slip and micro-rolling at first contact. This kinematic choice also decouples alignment from force transmission: in the approach phase the fingertip passively self-aligns, while the transmission remains mechanically disengaged via a lost-motion subassembly and therefore does not inject grasping force until engagement is intentionally triggered. The resulting grasp-then-lift sequencing reduces sensitivity to perception delays and small pose errors and simplifies low-level control.

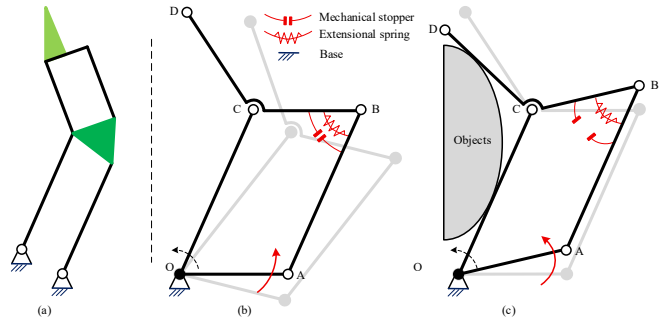

Fig. 2. Well-known finger mechanism having two phalanges constructed with the four-bar linkage. (a) Dual-Parallel Mechanism. (b) Four-bar linkage. (c) Shape Adaptation

B. Idle-Stroke Transmission for Grasp-to-Lift

To realize grasp-then-lift at the mechanism level, the hand places a lost-motion subassembly in series with a conventional four-bar. The subassembly comprises a driver-side lug, a mechanical stopper with a small gap δ to the lug, and a preloaded extension spring acting on link OA. Fig. 2b illustrates the idle-stroke phase: the input advances while the lug has not yet reached the stopper, torque into the four-bar is negligible, and the fingertips close and settle with minimal normal force. Fig. 2c shows the engaged phase: the lug contacts the stopper, the remaining input compresses the spring, and the four-bar generates the vertical output required for lifting.

This architecture brings two practical benefits. It decouples geometric alignment from force transmission, keeping interaction forces low until engagement is intentionally triggered, which reduces slip and sensitivity to small pose errors. It also removes the need for special four-bar ratios to obtain sufficient output stroke; the lost-motion element supplies the missing stroke while preserving a compact planar layout.

Key parameters are the idle gap δ, the spring stiffness k, the spring pre-load x_0, and the link set $\{OA, AB, BC, CO\}$The gap sets the timing of engagement. The product kx_0 governs the onset force and the available capacity. The link lengths define geometric leverage. In practice, δ is chosen slightly larger than the closure needed for stable contact, and kx_0 is selected to exceed the worst-case lift requirement with margin. The mechanism remains backdrivable during the idle stroke and becomes partially backdrivable after engagement, limited by the spring–stopper pair, which also provides a passive overload path. Section III details the resulting kinematic and quasi-static relations.

C. Idle-Stroke Gear: Symmetric Torsion-Spring Unit

In conventional four-bar lifters, geometric coupling around the base joint often causes the lower-left link to retreat during lifting, which back-constrains the upper branch. This makes it difficult to temporally separate pre-contact alignment from post-contact force build-up, leading to premature force peaks, wasted energy, and reduced robustness.

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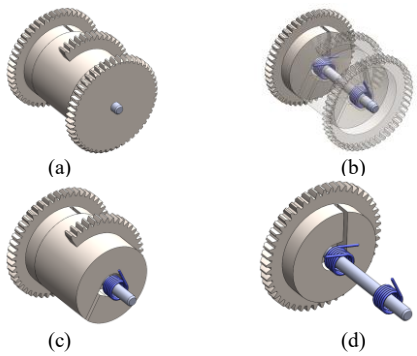

Fig. 3. The Idle-Stroke Gear. (a) Overall view of the idle-stroke transmission; (b) Overall perspective view; (c) Left-hand torsion spring; (d) Right-hand torsion spring.

An idle-stroke transmission is mounted on the upper-left long link, as shown in Fig. 3(a). The link is keyed to a spur gear to eliminate slip while permitting a small, bounded relative rotation within an idle-stroke window (Fig. 3(b)). Inside the housing, two torsion springs with identical stiffness are installed with opposite windings and opposite preloads (left-hand and right-hand), as illustrated in Fig. 3(c)–(d). This symmetric arrangement cancels bias torque, keeps the neutral position centered, and provides a compliant path during alignment. Within the idle-stroke window, the gear–link pair backdrives slightly to absorb coupling-induced retreat, so the output force rises only minimally. Once the relative motion exceeds the idle-stroke threshold, the spring pair produces a balanced restoring torque and the transmission engages to form a stable force path. Mechanical stops bound the maximum relative rotation and serve as an overload safeguard; beyond this limit the gear–link pair behaves as a rigid coupling. The mechanism preserves the original four-bar geometry while temporally separating alignment from force build-up, thereby reducing controller burden and enabling partial backdrivability.

III. Analysis

We model the dual-parallelogram fingertip transmission to obtain the mapping $\theta \to y_{\text{tip}}$, formalize the idle-stroke with the engagement angle θ_{eng}, and justify near-prismatic motion via a small rotation bound.

A. Kinematic

Consider the planar four-bar $\{OA, AB, BC, CO\}$. The input angle at the base joint is θ (at O) as shown in Fig. 4.

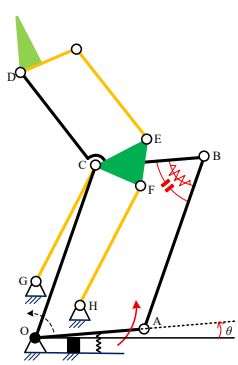

Fig. 4. Four-bar lifting unit with Idle-Stroke transmission.

Internal joint angles are φ (at A) and ψ (at C). A fixed frame {x, y} is attached at O with $\mathbf{e}_y$ along the fingertip normal. The fingertip position is $\mathbf{p}_{\text{tip}}(\theta,\phi,\psi)$, and its normal displacement is

$$y_{\text{tip}}(\theta) = \mathbf{e}_y^{\top}\mathbf{p}_{\text{tip}}\big(\theta,\phi(\theta),\psi(\theta)\big) \tag{1}$$

The four-bar loop closure is written in vector form

$$\mathbf{r}_{OA}(\theta) + \mathbf{r}_{AB}(\phi) = \mathbf{r}_{OC} + \mathbf{r}_{CB}(\psi) \tag{2}$$

from which $\varphi(\theta)$ and $\psi(\theta)$ are obtained by solving f (θ, φ, ψ) = 0. Substituting into (1) yields the kinematic map $\theta \to y_{\text{tip}}$. We define the tip Jacobian

$$J_{\text{tip}}(\theta) \triangleq \frac{d\,y_{\text{tip}}}{d\theta}. \tag{3}$$

The lost-motion unit introduces a thresholded effective input travel s(θ). Let θ_0 be the initial pose, see

$$s(\theta) = \max\{0,\, y_{\text{tip}}(\theta) - y_{\text{tip}}(\theta_0)\} \tag{4}$$

With a clearance $\delta > 0$, engagement occurs once $s(\theta) \geq \delta$, and we define the engagement angle

$$\theta_{\text{eng}} \triangleq \min\{\theta \mid s(\theta) \geq \delta\} \tag{5}$$

For $\theta < \theta_{eng}$ the transmission is idle; for $\theta \geq \theta_{eng}$ the geometric gain follows $J_{tip}(\theta)$.

Let $\alpha_{pad}(\theta)$ be the residual pad rotation from non-ideal parallelogram geometry. Linearizing about any operating point θ^* gives

$$\alpha_{\text{pad}}(\theta) \approx J_{\alpha}(\theta^*)(\theta - \theta^*) \tag{6}$$

and the bound

$$|\alpha_{\text{pad}}(\theta)| \leq \| J_{\alpha}(\theta^*)\| \, |\theta - \theta^*|. \tag{7}$$

With bar-parallelism tolerance e and coupler offset h relative to bar spacing d, we obtain

$$|\alpha_{\text{pad}}(\theta)| \leq \kappa e + \mathcal{O}(e^2), \qquad \kappa = \kappa\left(\frac{h}{d}\right); \tag{8}$$

choosing $h/d \ll 1$ keeps α_pad sub-degree over the usable stroke. We report $max_{\theta}\ |\alpha_{pad}(\theta)|$ and the RMSE between measured $y_{tip}(\theta)$ and the model

Let the admissible input range be $[\theta_{min}, \theta_{max}]$. To avoid geometric retreat prior to engagement and dead-zones, enforce monotonic approach

$$\frac{d\,y_{\text{tip}}}{d\theta} > 0, \qquad \forall \theta \in [\theta_{\min}, \theta_{\max}]. \tag{9}$$

Classical four-bar singularities occur when the mechanism Jacobian $J_{fb}(\theta)$ loses rank, i.e.,

$$\det\big(\mathbf{J}_{\text{fb}}(\theta)\big) = 0. \tag{10}$$

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Link ratios are selected so that θ_{eng} lies inside a non-singular, monotone sub-interval.

To expose scale-free trends, define the non-dimensional groups in

$$\bar{y} = \frac{y_{\text{tip}}}{l_{CO}}, \qquad \bar{\delta} = \frac{\delta}{l_{CO}}, \qquad \rho = \frac{l_{OA}}{l_{CO}}, \qquad \eta = \frac{l_{AB}}{l_{BC}}. \tag{11}$$

Numerically, $\partial\theta_{eng}/\partial\bar{\delta} > 0$, and there exists ρ^* that minimizes the curvature of $\bar{y}(\theta)$, yielding a flatter post-engagement gain $J_{tip}(\theta)$. In practice, choose (ρ, η) within bands that satisfy (9) while avoiding (10).

B. Quasi-Static Force Transmission

Under low-speed lifting with negligible inertia, model the spring as linear with stiffness k. The fingertip normal force is approximated by

$$F_{\text{tip}}(\theta) \approx k\, x_{\text{spring}}(\theta) J(\theta) = k\left(\max\{0, s(\theta) - \delta\} + x_0\right) J(\theta) \tag{12}$$

Where

$$J(\theta) \equiv \frac{\partial x_{\text{spring}}}{\partial y_{\text{tip}}} = \left(\frac{\partial x_{\text{spring}}}{\partial \theta}\right) \Big/ \left(\frac{\partial y_{\text{tip}}}{\partial \theta}\right) \tag{13}$$

is the spring-to-tip mechanical transmission Jacobian, and y_{tip} is the tip's normal displacement. To reflect the decoupling produced by the Idle-Stroke, use the effective definition

$$J(\theta) = \begin{cases} 0, & s(\theta) < \delta, \\ \dfrac{L_{\text{eff}}(\theta)}{\dfrac{\partial y_{\text{tip}}}{\partial \theta}}, & s(\theta) \geq \delta. \end{cases} \tag{14}$$

From (6) the behavior follows directly as shown in Fig. 5.

Idle: $s(\theta) < \delta$ gives $J(\theta) = 0$ and $F_{\text{tip}} \approx 0$, yielding translation with near-zero normal force.

Engagement: $s(\theta) \geq \delta$ produces controlled force build-up governed by k and $J(\theta)$.

If $J(\theta)$ is designed to be nearly flat over the working interval by tuning link lengths and pivot locations, the small-signal gain becomes

$$\frac{dF_{\text{tip}}}{ds} \approx k J(\theta) + k\left(\max\{0, s(\theta) - \delta\} + x_0\right) \frac{dJ}{ds} \tag{15}$$

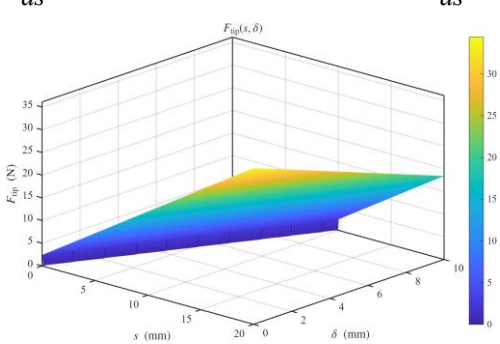

Fig. 5. Quasi-static force build-up across Idle and Engagement.

which is approximately constant and favors a stable force–displacement slope. Given a maximum usable travel s_{max}, a conservative peak-force bound is

$$F_{\max} \lesssim k \max\{0, x_0 + s_{\max} - \delta\} J_{\max} \tag{16}$$

which links (k, x_0, δ) and the geometric gain J to overload safety.

IV. Experiments

A. Setup

The prototype uses 3D-printed structural parts with standard bearings and fasteners. The drive side implements an adjustable idle-stroke, and the spring parameters follow the design section. The test bench offers interchangeable surface pads to vary friction, and objects include cubes, cylinders, thin plates, and items with distinct surface finishes. The gripper closes at a constant speed toward each object. A trial is counted as a success if the object is pinched and lifted to a fixed height without slip or drop within a short interval. Power is logged from motor voltage and current to estimate energy per cycle.

The evaluation targets feasibility rather than high-precision control. The gripper closes at constant speed toward the object. For $s(\theta) < \delta$ the system remains in Idle, providing a clear window for passive alignment with near-zero normal force. When $s(\theta) \geq \delta$ the force path closes, force builds smoothly, and the object is lifted. We used common items including small cubes, cylinders, thin plates, and parts with distinct surface frictions, spanning typical sizes and masses for small-scale manipulation. A trial is counted as a success if the object is stably pinched and raised to a set height without slip or drop within a short interval.

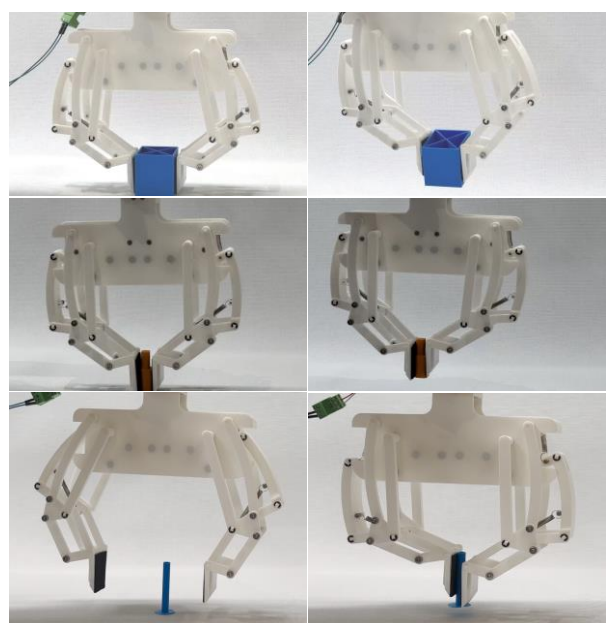

Fig. 6. Prototype and feasibility demonstration.

The Fig. 6 sequence shows that the Idle window preserves alignment motion without premature pushing, crossing δ triggers engagement and a gentle force ramp, and lifting

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proceeds with low controller burden. For irregular edges and thin plates the Idle-Stroke suppresses early hard contact and sliding. Under mild disturbance the mechanism exhibits partial backdrivability and quickly returns to the loaded state, consistent with the analysis. Limitations stem from rapid-prototyped materials and assembly tolerances, which bound the performance under high load and long duty. On very soft or adhesive surfaces a small amount of drag can appear during Idle; reducing δ or using lower-friction fingertip covers mitigates this. Overall, the observations align with the model and support the feasibility and applicability of the Idle-Stroke transmission on real hardware.

TABLE I. IDLE-STROKE VS. NO-IDLE ACROSS LINK-LENGTH RATIOS

Ratio set	Success (IS)[%]	Success (NIS) [%]
1:2:2.4:2	100	100
1:2:2:2	100	0
1:2:2.5:2	100	0

To isolate the effect of the idle-stroke from geometry, we first fix the four-bar link-length ratio and toggle the transmission between idle-stroke enabled and no-idle (mechanically locked) while keeping hardware and control identical. We then sweep several nearby ratio sets around the nominal design. As summarized in Table I, with the idle-stroke enabled, all tested ratio sets complete grasp-and-lift robustly. In contrast, without the idle-stroke, only a single special ratio achieves lifting; the others fail due to premature force build-up and misalignment during the alignment-to-lift transition.

V. CONCLUSION

This work introduced and validated GLINT, a grasp–lift mechanism that embeds grasp-to-lift phase switching into an idle-stroke transmission, enabling a smooth transition from near force-free alignment to stable, load-bearing lifting with a single actuator. We formulated a compact kinematic and quasi-static model that specifies the engagement condition and the tip force–displacement relationship, and we distilled these results into actionable design guidelines for the key parameters. The proposed formulation clarifies how structural choices shape the onset of engagement, the amplification of contact forces, and the avoidance of undesirable singularities, thereby offering a principled route to controller-light grasp–lift behaviors.

A single-DoF prototype demonstrated that the mechanism maintains low structural and control complexity while achieving reliable phase switching across object shapes and surface frictions. The experiments further indicate that the idle-stroke window can tolerate modest modeling error and manufacturing variability while preserving predictable switching and force transmission. Together, the analysis and empirical evidence suggest that GLINT is a practical building block for resource-constrained manipulation where hardware simplicity, energy efficiency, and robustness to perception or actuation imperfections are prioritized.

This study also exposes several avenues for growth. First, the present characterization of contact conditions does not yet cover extreme friction regimes or complex surface geometries; extending the model and tests to those cases is a direct next step. Second, long-horizon durability and energy statistics merit broader sampling to assess wear, drift, and maintenance requirements. Third, scaling the concept to multi-DoF end-effectors and more demanding tasks—such as in-hand reconfiguration, non-prehensile transitions, and interaction under motion—requires integrating minimal sensing with adaptive control and task-level optimization. Looking ahead, we plan to explore these directions and evaluate deployment in warehouse picking, mobile and aerial manipulation, and multi-finger hands, aiming to turn GLINT into a general-purpose, low-overhead pathway to reliable grasp-to-lift performance.

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